

Data Quality & Discrepancy Detection System

Generalised — works on any dataset

SECTION 1 — DISCREPANCY TYPES

Discrepancy	Crit.	AI Level	Example	How Detected	AI Usage & Cost
Missing / Null Values Empty cells, blanks, None or placeholder strings like 'N/A'.	High	Rules Only	Price column: 340 rows have no value at all.	Count nulls per column; flag if null% > 5% (Medium) or > 30% (High).	No AI required.
Wrong Data Type Value doesn't match the column's expected type.	High	Rules Only	Age column contains 'twenty-five' instead of 25.	Infer expected type from majority of values; flag rows that can't be cast.	No AI required.
Duplicate Records Two or more rows representing the same real entity.	Medium	Partial AI	Supplier 'ABC Steels' appears twice with slightly different phones.	Hash rows for exact dupes; fuzzy-match name/address for near-dupes.	Rules find candidates. AI only resolves ambiguous near-duplicate pairs: 'Which row is authoritative?' Very low cost — called per pair, not per row.
Format Inconsistency Same value written in multiple formats across rows.	Medium	Rules Only	Dates as 01/02/2024 in some rows and 2024-01-02 in others.	Detect dominant format per column via regex; flag outlier rows.	No AI required.
Out-of-Range Values Numeric value outside logical or statistical bounds.	High	Partial AI	Employee age recorded as 312. Order value 9,80,00,000 vs typical 5,000–50,000.	Config-defined logical limits + IQR / Z-score statistical detection.	Rules flag all out-of-range rows. AI is called only for borderline cases to assess: genuine extreme or entry error? Not called for clear violations.
Whitespace & Encoding Errors Leading/trailing spaces, double spaces, or garbled characters.	Low	Rules Only	City stored as ' Delhi' (leading space); name shows 'St��el' due to encoding.	Compare value to its stripped form; detect non-ASCII in ASCII-expected columns.	No AI required.
Inconsistent Casing Same value written in different capitalizations in the same column.	Low	Rules Only	'mumbai', 'Mumbai', 'MUMBAI' all in the City column.	Lowercase-group all values; report variant counts; flag collision columns.	No AI required.
Unit Mismatch Numeric column has values in two different units — both look valid.	High	Full AI	Weight column: 55, 70, 80 (kg) mixed with 121, 154, 176 (lbs).	Detect bimodal distribution; check adjacent columns for unit hints.	AI always used: receives sample rows + column context, identifies the unit split, and outputs a classification rule. Rule runs locally on all rows — one AI call per suspect column.
Impossible Combinations Individual values are valid but their combination is logically impossible.	High	Full AI	Shipment dispatch date is earlier than the order date.	Apply config-defined cross-column rules first; AI extends coverage.	AI always used: gets column names + 100 sample rows, returns domain-agnostic validation rules as JSON. Rules cached and reused on future runs — one AI call per dataset schema.
Statistical Outliers Technically valid values that are statistically extreme for their column.	Medium	Partial AI	All invoices are 5,000–50,000. One is 4,87,00,000 — passes all other checks.	Z-score and IQR flagging per column; Isolation Forest for multi-column outliers.	Rules flag all statistical outliers. AI is called for ambiguous ones to return a verdict: likely error / plausible extreme / uncertain. Clear extremes (e.g. 3000% above mean) are flagged without AI.

Discrepancy	Crit.	AI Level	Example	How Detected	AI Usage & Cost
Free-Text Normalization Text column with many variations of the same value.	Medium	Full AI	'Sr. Software Developer', 'Senior Software Dev', 'Sr Dev' — same role, 3 buckets.	Detect high-cardinality text columns; sample up to 200 unique values.	AI always used: groups sampled unique values into canonical categories and returns a mapping dict. Mapping applied to full column locally — one AI call per column, regardless of row count.
Schema Drift Column added, removed, or renamed vs last known structure.	High	Rules Only	Last month's file had 'GST Number'. This month it is absent.	Compare current column list against stored baseline in config. Runs first, before all agents.	No AI required.
Row Count Anomaly File has significantly more or fewer rows than expected.	High	Rules Only	Yesterday: 12,400 rows. Today: 7,800 rows — no explanation.	Track row counts across runs; flag if change exceeds threshold (e.g. 20%).	No AI required.
Sequence / Temporal Ordering Events out of logical order, or sequential IDs with unexplained gaps.	High	Rules Only	Customer return recorded before their purchase. Invoice jumps 1042 to 1047.	Check date-field ordering within groups; detect gaps in ID sequences.	No AI required.
Aggregate Mismatch A stated total does not match the sum of its detail rows.	High	Rules Only	Invoice header: 24,500. Line items sum: 23,800.	Recalculate expected totals from detail rows; flag any discrepancy.	No AI required.
PII in Wrong Column Sensitive identifiers found in a column not designed for them.	High	Partial AI	An Aadhaar-format number in the free-text 'Notes' field.	Apply PII regex patterns (Aadhaar, PAN, phone) to non-designated columns.	Rules catch clear pattern matches. AI handles partial or ambiguous matches that don't fully satisfy a regex but look suspicious in context.

- **Rules Only** — No AI ever called. Fully deterministic logic runs on the complete dataset.
- **Partial AI** — Rules run first and handle most cases. AI called only when rules leave genuine ambiguity.
- **Full AI** — AI is always invoked for this agent — but only on a small sample (100–200 rows), never the full dataset.

SECTION 2 — SYSTEM ARCHITECTURE

The system is generalised — it works on any uploaded dataset without prior knowledge of its domain. The architecture has five stages, designed so that AI is used sparingly and cost is always bounded.

1	Upload	User uploads a CSV or Excel file via the Streamlit browser interface.
2	Profile	System scans the full dataset locally: row count, column names, data types, null%, min/max, top-N unique values. No AI, no cost.
3	AI Column Classification	Column names + data types + basic stats are sent to the AI in a single call. AI decides which agents are relevant for each column (e.g. 'free-text normalization applies to job_title, not to invoice_id'). This prevents every agent running blindly on every column.
4	Parallel Agents	All applicable agents run simultaneously — like 16 inspection lanes open at once. Rule-based agents finish in seconds. AI agents send only small samples (100–200 rows), not the full file.
5	Rank & Display	All findings are merged, ranked by criticality (High first) then by rows affected, and displayed on the Streamlit dashboard with column, issue type, example value, and suggested interpretation.

Key design: AI classifies columns before any agent runs. The system sends only column names, inferred types, and summary stats to the AI — never the full data. The AI returns a JSON map: {column_name: [applicable_agents]}. This means agents like 'Free-Text Normalization' only run on columns where they make sense (e.g. job_title, city, category) and are skipped for columns like invoice_id or timestamp. This single upfront AI call controls scope for all subsequent agents.

Parallel execution. All applicable agents run simultaneously — similar to an Airflow DAG where tasks with no dependency on each other are dispatched at the same time. Rule-based agents complete in milliseconds. AI agents each make one small API call independently. Results from all agents are collected, merged, and ranked when the last agent finishes.

SECTION 3 — TOOLS REQUIRED

Tool	Purpose	Cost
Python 3.10+	Core engine — runs all agents and profiling.	Free
Streamlit	Browser-based dashboard. No coding to operate.	Free
VS Code / Cursor	Used once to start the system. Semi-tech friendly.	Free
Anthropic API Key	Required for AI agents only. Pay-per-use.	~\$40–80 / run*
Git (optional)	Sync project across devices. Not needed daily.	Free

* AI cost estimate for a typical dataset of 50,000–5,00,000 rows. Subsequent runs on the same schema cost significantly less as AI-generated rules are cached in config.yaml and reused without additional API calls.